

NYC AIRBNB LISTING INTELLIGENCE

(2025 – 2026)

Data Analysis Project Report
by
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Prepared as a Power BI / Data Analytics Portfolio Project

Source: fact_listing and 7 dimension tables (gold layer) · 4-page Power BI dashboard (Overview, Host, Room, Rating)

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1. Executive Summary

This report documents a completed Power BI analytics project on a curated (“gold layer”) NYC Airbnb dataset (2015–2026): one fact table and seven dimension tables (host, location, room, bathroom, availability, review, date) describing **30,260** listings across five boroughs, visualized in a four-page dashboard (Overview, Host, Room, Rating).

At the marketplace level: Total Estimated Revenue of **389M**, Revenue per Available listing of **18.11K**, **30K** total listings of which **22K** are active, average occupancy of **14.4%**, and an average guest rating of **4.77/5**. Manhattan drives **52.78%** of revenue (**205M**) on an average daily rate of **\$366** — over **70%** above Brooklyn's **\$213**.

Host-side: **16,442** hosts averaging **4.90** listings each, a **19.8%** Superhost rate, and **98.9%** identity verification. Room-level: Entire home/apt dominates supply (**55%**) but has the lowest availability, while Hotel room — a small niche — commands the highest rate (**\$662.4**) and highest availability. Ratings stay consistently strong (**4.64–4.84** across sub-categories).

Collectively, the dashboard turns a normalized, multi-table dataset into a decision-ready BI tool for marketplace strategy, host performance, pricing, and guest-experience tracking.

2. Project Introduction

2.1 Background & Business Problem

Airbnb's NYC marketplace spans wide variation in borough-level pricing, room configurations, host portfolio sizes, and availability. NYC listing data is typically fragmented across host, property, bathroom, availability, review, and pricing attributes, making it hard to answer basic marketplace-management questions — which boroughs/room types generate the most revenue, which hosts control supply, and whether guest satisfaction is consistent. This project consolidates those sources into a star-schema model and four-page Power BI dashboard for monitoring and acting on marketplace performance.

2.2 Objectives & Key Business Questions

- Understand how listings, revenue, and pricing are distributed across boroughs and room types, and which boroughs generate the highest revenue share.
- Evaluate host-side supply concentration — portfolio size, tenure, verification, and Superhost status.
- Analyze pricing patterns by borough, room type, and bathroom privacy/category.
- Compare room-type performance on availability, capacity, and bed configuration.
- Study guest review activity and rating consistency across the marketplace.

3. Dataset Overview

3.1 Data Source

The project uses a curated “gold layer” extract (eight Excel workbooks) representing a processed, analytics-ready view of an NYC Airbnb listings dataset for 2025–2026. Each workbook corresponds to one table in the data model, and every table shares the same 30,260-row grain, keyed by ListingKey. The specific upstream raw source (e.g., Inside Airbnb or a similar public NYC listings export) was not stated in the provided materials, so it is not asserted here.

3.2 Dataset Structure

The gold layer consists of one fact table and seven dimension tables, all joined on the common surrogate key ListingKey. This is a listing-grain star schema: every dimension table carries exactly one row per listing rather than one row per host or per booking event, which is an important modeling characteristic discussed further in Section 4.

3.3 Fact and Dimension Tables

Table	Grain	Purpose
gold.fact_listing	1 row / ListingKey	Core listing-level measures: price, estimated revenue, occupancy, availability windows, review counts, and rating sub-scores.
gold_dim_host	1 row / ListingKey	Host attributes attached to the listing: host ID/name, Superhost and identity-verification flags, host's total/active/room-type listing counts, and host tenure (user years/months, host years/months).
gold_dim_location	1 row / ListingKey	Borough, neighborhood, and latitude/longitude of the listing.
gold_dim_room	1 row / ListingKey	Room type, capacity, bedrooms, beds, an estimated-beds figure, and a bed-data-quality flag (Original / Estimated / Adjusted).
gold_dim_bathroom	1 row / ListingKey	Bathroom type text, bathroom count, privacy (Private/Shared/Unknown), and a bathroom category.
gold_dim_avability	1 row / ListingKey	HasAvailability flag and AvailYearEnd indicator.
gold_dim_review	1 row / ListingKey	FirstReview and LastReview dates for the listing.
gold_dim_date	1 row / ListingKey	CheckInDate and CheckOutDate for the listing's price-quote window (confirmed below).

Correction based on the supplied SQL: gold.dim_date is built directly from price_quote_checkin_date and price_quote_checkout_date in the Silver table (see gold_stored_procedure.sql). These are the check-in/check-out dates used when the source scrape requested a live price quote for each listing — a single simulated future stay window — not a historical record of actual guest bookings or stays. Any dashboard use of dim_date should therefore be read as “the stay window priced at time of data collection,” not as booking history.

3.4 Data Granularity

All eight tables contain 30,260 data rows, confirming a one-row-per-listing grain throughout the model — this matches the dashboard's Total Listing KPI of 30K. Because gold_dim_host also carries one row per ListingKey (rather than one row per unique host), a host with multiple listings appears multiple times in that table, once per listing, each time repeating that host's portfolio-level attributes (TotalListings, ActiveListings, tenure, etc.). This is confirmed directly from the column headers and sample rows of the workbook and is treated as a modeling fact rather than an assumption.

3.5 Important Fields

Field	Table	Description
ListingKey	All tables	Surrogate key joining fact and dimension tables (1 row per listing).
Price / EstRevenue / EstOccupancy	fact_listing	Listing price, estimated revenue, and estimated occupancy.
Avail30/60/90/365Days	fact_listing	Days available within each rolling window.
TotalReviews / ReviewsPerMonth / Rating sub-scores	fact_listing	Review volume, review frequency, and Accuracy/Cleanliness/CheckIn/Communication/Location/Value ratings.
Borough / Neighborhood	gold_dim_location	Geographic classification used for all borough-level cuts.
RoomType / Capacity / Bedrooms / Beds	gold_dim_room	Physical configuration of the listing.
BathroomPrivacy / BathroomCategory	gold_dim_bathroom	Whether the bathroom is private, shared, or unknown, and its category.
IsSuperhost / IsIdentityVerified / TotalListings / HostYears	gold_dim_host	Host status, verification, portfolio size, and tenure.

4. Data Architecture & Data Model

The project implements a three-layer SQL Server warehouse (bronze → silver → gold), confirmed directly from the provided DDL and stored-procedure scripts, followed by a Power Query (M) enrichment step and a DAX measure layer inside Power BI:

CSV Source Files → dbo (staging) → Bronze (raw, NVARCHAR) → Silver (typed & cleaned) → Gold (star-schema views) → Power Query M (enrichment & host dedup) → Power BI Data Model → DAX Measures → Dashboard

Layer-by-Layer Architecture (confirmed from SQL_Scripts/)

- **dbo (staging):** raw CSV files are imported into SQL Server into a dbo.nyc_airbnb_listing staging table, per the project owner's description and the bronze.load_bronze procedure's source reference.

- Bronze (bronze.NYC_airbnb_listing): every column is stored as NVARCHAR(100) via TRY_CAST from the dbo staging table (bronze_listing_ddl.sql, bronze_nyc_airbnb_listing_stored_procedure.sql). This preserves raw values with no type enforcement, acting as an auditable raw layer.
- Silver (silver.NYC_airbnb_listing): silver.load_silver truncates and reloads the table from Bronze, applying TRY_CAST/TRY_CONVERT with NULLIF for null handling, TRIM for text fields, and several derived columns (detailed with exact logic in Section 5).
- Gold (gold schema views): gold_stored_procedure.sql defines one view per dimension (dim_host, dim_location, dim_room, dim_bathroom, dim_availability, dim_review, dim_date) plus fact_listing, all selecting from silver.NYC_airbnb_listing and renaming/deriving business-friendly columns (e.g., host_is_superhost = 't' → IsSuperhost = 1).
- Power Query (M) layer: power_query_m_code.m shows that gold.dim_host is further transformed after landing in Power BI — adding “Superhost Status” and “Identity Verification Status” text columns, a “Host portfolio” category, and, critically, de-duplicating the table down to one row per unique HostID (Table.Distinct applied on a derived “Unique Host” column and again on HostID).
- DAX measure layer: dax_measures_columns_script.txt defines ~35 measures and 2 calculated columns spanning availability, revenue, host, and rating analysis (full list in Section 19.2), consumed by the four dashboard pages.

Star Schema Structure — Confirmed from the Power BI Model Diagram

The project owner has since provided a screenshot of the actual Power BI model view, reproduced as Figure 1 below. This is the authoritative source for relationship structure and supersedes the inference made from the Power Query script in an earlier draft of this report.



Figure 1. Power BI model view — gold.fact_listing related 1:1 to all seven gold dimension views on ListingKey, plus two disconnected calculated tables (Key Measures, Availability Horizon) and a hidden Measure_Table used to organize DAX measures.

- Confirmed cardinality: every one of the seven dimension tables — `dim_host`, `dim_review`, `dim_bathroom`, `dim_room`, `dim_location`, `dim_availability`, and `dim_date` — shows a “1” cardinality marker on both ends of its relationship to `gold_fact_listing`. All seven relationships are 1:1 on `ListingKey`, with no exceptions.
- Correction to the previous draft: this means `gold_dim_host` is listing-grain in the actual working model (30,260 rows, matching every other table), not de-duplicated to host-grain. The earlier explanation — that Power Query's `Table.Distinct` steps reduced `dim_host` to one row per unique host before reaching the model — does not hold up against this diagram and is retracted. Whatever de-duplication logic exists in the M script, it is not reflected in the relationship structure actually built and used by the dashboard.
- Practical implication: with `dim_host` confirmed listing-grain and related 1:1 to `fact_listing`, a plain Sum aggregation of `TotalListings` across a matrix visual (such as the Host page's “Top 10 Host by Listing” table, grouped by Borough) will add that host's `TotalListings` value once for every NYC listing they own, inflating the true sum. Distinct-count-based measures, like the DAX script's `Total Host count` (`DISTINCTCOUNT(HostID)`), are unaffected by this because `DISTINCTCOUNT` de-duplicates at query time regardless of the underlying table's grain — which is exactly why `Total Host` (16.442K) is reliable while a raw Sum of `TotalListings` by Borough is not.
- Cross-filter direction: the `dim_room` relationship is visibly configured with bidirectional (“Both”) cross-filtering, shown by a distinct double-headed arrow on its connector — the only relationship drawn this way among the seven. The rest appear to use the tool's default single-direction filtering (dimension filters fact).
- Confirmed calculation objects: `Key Measures` (a calculated table equal to `DISTINCT(fact_listing[ListingKey])`) and `Availability Horizon` (a manually authored `DATATABLE`) both appear in the diagram exactly as defined in `dax_measures_columns_script.txt`, and a hidden `Measure_Table` serves purely as an organizational home for DAX measures — all confirmed, good-practice modeling choices.
- Hidden keys: every `ListingKey` field (and `HostID` in `dim_host`) is marked hidden in the diagram, preventing report users from browsing raw join keys in the field list — a standard modeling hygiene practice.

This is a case where new evidence changed the conclusion. It's flagged explicitly, and the affected sections below (9.2 and 16) have been updated to match, rather than leaving the earlier, now-unsupported explanation in place.

5. Data Cleaning & Preparation

With the SQL scripts now available, the cleaning steps below are confirmed directly from `silver.load_silver` and the gold-layer view definitions, rather than inferred from field names.

Bronze → Silver (`silver.load_silver`)

- Null handling: virtually every column uses `TRY_CAST(NULLIF(column, ''), AS <type>)` or `TRY_CONVERT(DATE, NULLIF(column, ''), 105)`, converting empty strings from the raw Bronze layer into true NULLs before type casting, and safely handling any values that fail to convert (rather than erroring).
- Surrogate key generation: `ListingKey` is generated with `ROW_NUMBER() OVER (ORDER BY id, host_id)`, creating a stable, ordered integer key used across the entire gold layer in place of the platform's native `id`.
- Text cleaning: `TRIM(NULLIF(...))` is applied to `host_name`, `neighbourhood_cleansed`, `neighbourhood_group_cleansed`, and `room_type` to remove stray whitespace.
- Price cleaning: `price` is cleaned with `TRY_CAST(REPLACE(REPLACE(NULLIF(price, ''), '$', ''), ',', '')) AS DECIMAL(15,2))` — stripping currency symbols and thousands separators before casting to a numeric type.

- Host tenure derivation: `host_as_user_total_months` and `host_as_host_total_months` are computed as $(\text{years} \times 12) + \text{months}$, using `ISNULL` to treat a missing year or month component as 0 while still returning `NULL` when both components are missing — this is the exact source of `TotalUserMonths / TotalHostMonths` in `gold_dim_host` and, downstream, the dashboard's Host Tenure Category segmentation.
- Bathroom text parsing: `bathrooms_count` and `bathroomtype` are derived from the raw `bathrooms_text` field using a pattern check (`bathrooms_text LIKE '[0-9]%' AND CHARINDEX(' ', bathrooms_text) > 0`). Where the pattern matches, the leading number becomes `bathrooms_count` and the remaining text becomes `bathroomtype`; where `bathrooms_text` is `NULL` or blank, `bathroomtype` is explicitly set to 'Unknown'.
- Precision standardization: latitude/longitude are rounded to 6 decimal places and `review_scores_rating` is rounded to 1 decimal place with `ROUND()`, standardizing precision across the dataset.

Silver → Gold (gold_stored_procedure.sql)

- Boolean standardization: `host_is_superhost`, `host_identity_verified`, and `has_availability` (stored as 't'/'f' text flags in Silver) are converted to 1/0 integer flags (`IsSuperhost`, `IsIdentityVerified`, `HasAvailability`) in the gold views — explaining why the dashboard can show these as clean percentages (e.g., Super Host Rate 19.8%).
- Beds imputation with a quality flag: `gold_dim_room` sets `EstimatedBeds = bedrooms` when `beds` is `NULL`, or when `beds < bedrooms` (treating a bed count lower than the bedroom count as unreliable), otherwise keeping the original `beds` value. `BedDataQuality` is set to 'Estimated' when `beds` was `NULL`, 'Adjusted' when `beds < bedrooms`, and 'Original' otherwise — this is the exact logic behind the Beds Data Quality visual in Section 9.3.
- Bathroom categorization: `gold_dim_bathroom` classifies `BathroomPrivacy` as 'Private' when `bathroomtype` contains "Private", 'Shared' when it contains "Shared", and 'Unknown' otherwise — meaning the Unknown category (Section 9.1) is specifically the set of listings whose bathroom text does not literally contain the word "Private" or "Shared" (e.g., a plain "1 bath" entry), rather than a separate collection process. `BathroomCategory` similarly classifies text containing "Half" as 'Half', text containing "private bath"/"Shared bath"/"shared baths" as 'Full', and everything else as 'Unknown'.
- Rating rounding: `fact_listing` casts and rounds each rating sub-score to one decimal place, e.g., `CAST(ROUND(review_scores_rating,1) AS DECIMAL(2,1)) AS Rating`.

Power Query (M) enrichment, applied after loading into Power BI

- Descriptive labels: adds "Superhost Status" (Yes/No/Unknown) and "Identity Verification Status" (Verified/Not Verified/N/A) text columns derived from the 0/1 flags, for direct use in visuals and slicers.
- Host de-duplication: applies `Table.Distinct` on a derived "Unique Host" column (equal to `HostID`) and again directly on `HostID`, collapsing the listing-grain `gold.dim_host` view down to one row per unique host before it is used in the Power BI model (see Section 4 for the modeling implications of this step).
- Host portfolio segmentation: adds a "Host portfolio" column using fixed `TotalListings` thresholds — 1 listing = "Single-Listing Host", 2–5 = "Small Portfolio", 6–20 = "Medium Portfolio", 21+ = "Large Portfolio", and `NULL` = "N/A" — this is the exact logic behind the Host Portfolio donut in Section 9.2.

No explicit evidence of outlier treatment (e.g., capping extreme prices) was found in the provided scripts, so the outlier listings visible in the Overview page's scatter plot (Section 9.1) appear to remain untreated in the gold layer.

6. Analytical Methodology

The analysis moves from a normalized, listing-grain gold layer to business-ready insight through the following stages:

- SQL-based ETL (SQL Server): a bronze → silver → gold pipeline, implemented as stored procedures and views, performs the actual data cleaning, typing, and dimensional restructuring before the data ever reaches Power BI (Sections 4–5).
- Power Query (M) enrichment: a further transformation pass inside Power BI adds descriptive status columns, host portfolio segmentation, and de-duplicates the host table (Section 5).
- Data modeling in Power BI: the gold tables are loaded and related on ListingKey (dim_host related at host grain post-dedup, all other dimensions at listing grain), enabling cross-filtering between any dimension (borough, room type, bathroom privacy, host tenure) and the fact measures (price, revenue, occupancy, reviews, ratings).
- DAX measure layer: 33 measures and 2 calculated columns (dax_measures_columns_script.txt) compute every KPI card and calculated category (Host Tenure Category, Listing Status) shown on the four dashboard pages — the full list is reproduced in Section 19.2.
- Exploratory and comparative analysis: the four dashboard pages perform borough-vs-borough, room-type-vs-room-type, and host-tenure-vs-tenure comparisons using bar, donut, ribbon, and scatter visuals.
- Ranking analysis: “Top 10 Host by Listing” and “Top 1000 Revenue Listing” tables apply ranking logic to surface the highest-impact hosts and listings.
- Segmentation: hosts are segmented by tenure (New/Emerging/Established/Experienced/Long-Term/Veteran) and by portfolio size (Single-Listing/Small/Medium/Large Portfolio); listings are segmented by Review Activity Status (Active/Slowing/Low Activity/Potentially Inactive/No Review Data).

This reflects a standard BI workflow — data modeling and DAX doing the heavy computational lifting, with Power BI visuals used for exploratory and comparative storytelling — rather than predictive or statistical modeling, which is outside the scope of what is shown.

7. KPI Framework

KPI	Value	Business Meaning	Dashboard Page
Total Estimated Revenue	389M	Aggregate projected revenue across all listings; the top-line marketplace-size metric.	Overview
Revenue per Available (listing)	18.11K	Average estimated revenue normalized per available listing; a per-unit productivity measure.	Overview
Total Listing	30K	Total number of listings in the dataset (30,260 confirmed from row counts).	Overview
Active Listing	22K	Subset of listings currently marked as active/available for booking.	Overview
Avg Occupancy	14.4%	Average estimated occupancy rate across all listings.	Overview

KPI	Value	Business Meaning	Dashboard Page
Overall Rating	4.77	Average guest rating across all listings; DAX confirms this is CALCULATE(AVERAGE(Rating), TotalReviews >= 5) — i.e., filtered to listings with at least 5 reviews (measure: Avg Overall Rating).	Overview / Rating
Total Host	16.442K	Number of distinct hosts represented in the data (DISTINCTCOUNT(HostID), measure: Total Host count).	Host
TotalListings (Host page)	16.353K	Listings tied to hosts as counted on this page. The exact card is not individually named in the provided DAX script; the closest candidates are an implicit COUNT of the (now host-deduplicated, per Section 4) TotalListings column, which would be slightly below Total Host if a small number of hosts have a blank TotalListings value.	Host
Super Host Rate	19.8%	Share of hosts holding Superhost status.	Host
Host Active Listings	16.0%	Share of host portfolios currently active.	Host
Identity Verification	98.9%	Share of hosts with verified identity — a trust/safety indicator.	Host
Avg Listings per Host	4.90	Average portfolio size per host — an indicator of supply concentration.	Host
Total Availability	30K	Count of listings with availability data captured.	Room
Avg Availability	101.07	Average number of available days (window not separately labeled on the KPI card).	Room
Avg Capacity	2.86	Average guest capacity per listing.	Room
Beds per Bedroom	1.47	Average bed-to-bedroom ratio, a space-utilization indicator.	Room
Avg Occupancy (Room page)	14.36%	Occupancy restated on the Room page, consistent with the Overview figure.	Room
Average of Estimated Beds	1.69	Average imputed/estimated bed count per listing.	Room
Overall Rating / Value / Location / Communication / Cleanliness / Check-In Ratings	4.77 / 4.64 / 4.75 / 4.83 / 4.68 / 4.84	Sub-category guest satisfaction scores.	Rating
Avg of Total (reviews measure)	32.41	Average of a review-related measure per listing (exact underlying field label was truncated on the card).	Rating

Resolved — Rating discrepancy explained by DAX: the Rating page's three closely related figures are now fully explained by the provided DAX. The “Overall Rating” KPI card (4.77) uses the Avg Overall Rating measure,

CALCULATE(AVERAGE('gold fact_listing'[Rating]), TotalReviews >= 5) — a direct average of the Rating field, restricted to listings with at least 5 reviews. The “Rating Across Categories” gauge (4.75) uses a different measure, Average Rating Across Categories, which computes AVERAGEX across all listings (no review-count filter) of the mean of the six sub-category ratings (Value, Accuracy, CheckIn, Cleanliness, Communication, Location) — a compositional average that excludes the Rating field itself and excludes no listings by review volume. The “Overall Rating” gauge (4.73) is most likely an unfiltered AVERAGE('gold fact_listing'[Rating]) with no review-count threshold, which would sit slightly below the >=5-reviews-filtered 4.77 KPI if lower-reviewed listings skew marginally lower. All three values are legitimate, differently-scoped measures rather than an error, and are documented in full in Section 19.2.

DAX

No DAX formulas were included in the provided project materials. The dashboard clearly uses DAX-based measures to power every KPI card and calculated category (Host Tenure Category, Review Activity Status, Bathroom Category, Beds Data Quality); the exact formulas were not provided and are therefore not fabricated here.

8. Power BI Dashboard Overview

Overview Page

Element	Details
Purpose	Marketplace-level summary of revenue, pricing, and listing volume.
Main KPIs	Total Estimated Revenue, Revenue per Available, Total Listing, Active Listing, Avg Occupancy, Overall Rating.
Main Visualizations	Revenue by Borough (donut), Avg Daily Rate by Borough, Avg Daily Rate by RoomType, Avg Daily Rate by Bathroom Privacy, Listing Price vs Beds & Capacity (scatter), Top 1000 Revenue Listing table.
Filters / Slicers	Capacity, Price, Overall Rating, Listing Status, Superhost Status, Borough/Neighborhood, RoomType, BathroomCategory, BathroomPrivacy (global slicer panel, present on all pages).
Questions Answered	Where is revenue concentrated? How does price vary by borough, room type, and bathroom privacy? Which listings drive the most revenue?
Business Value	Gives leadership a single-glance view of marketplace size and where commercial value is concentrated.

Host Page

Element	Details
Purpose	Analysis of host supply concentration, tenure, and portfolio structure.
Main KPIs	Total Host, TotalListings, Super Host Rate, Host Active Listings, Identity Verification, Avg Listings per Host.

Element	Details
Main Visualizations	Top 10 Host Performance by Total Listing, Daily Rate vs Price Value Rating, Revenue by Borough and Host Tenure Category, Host Tenure Category by Super Host, Host Tenure Category by Room Type, Host Portfolio (donut), Top 10 Host by Listing (table by Borough).
Filters / Slicers	Same global slicer panel as Overview.
Questions Answered	How concentrated is host supply? What share of hosts are Superhosts or verified? How does tenure relate to revenue and room-type mix?
Business Value	Supports host-relationship management, retention strategy, and identification of high-value/professional hosts.

Room Page

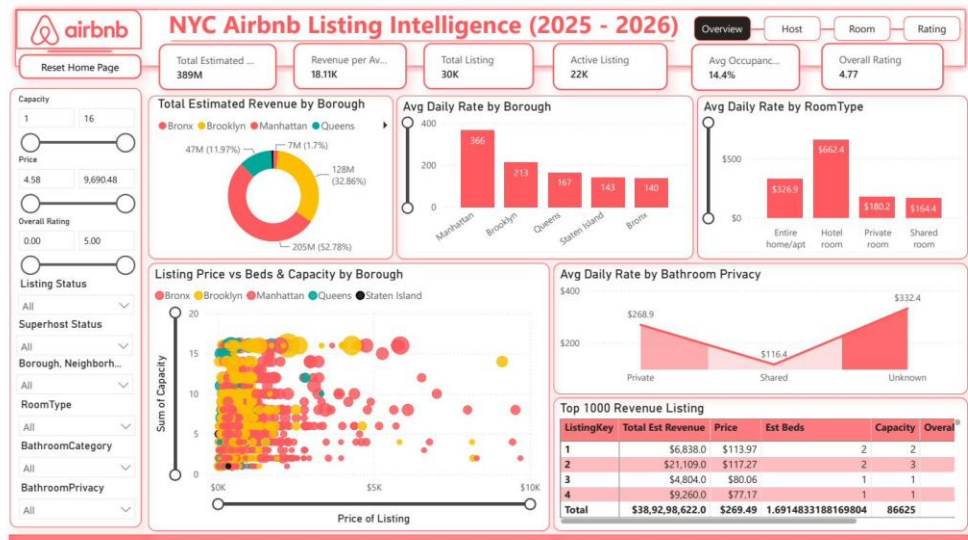
Element	Details
Purpose	Analysis of room type, capacity, bathroom configuration, and availability.
Main KPIs	Total Availability, Avg Availability, Avg Capacity, Beds per Bedroom, Avg Occupancy, Average of Estimated Beds.
Main Visualizations	Availability Rates by Borough (30–365 days), Availability by Room Type, Room Type vs Bathroom Privacy, Average Bed & Capacity by Room Type, Listing vs Borough & Room Type, Total Listing by RoomType (donut), Beds-to-Bedroom Ratio by Room Type (donut), Beds Data Quality, Total Bathrooms by Room Type (donut).
Filters / Slicers	Same global slicer panel as Overview.
Questions Answered	How does availability differ by borough and room type? How do bed/bathroom configurations vary by room type? How reliable is the underlying bed count data?
Business Value	Helps evaluate supply utilization and configuration quality — relevant to both host guidance and marketplace supply strategy.

Rating Page

Element	Details
Purpose	Analysis of guest satisfaction and review engagement.
Main KPIs	Overall Rating, Value Rating, Location Rating, Communication Rating, Cleanliness Rating, Check-In Rating, Avg of Total (review measure).
Main Visualizations	Rating Across Categories (gauge), Overall Rating (gauge), Count of ListingKey by Borough and Review Activity Status, Daily Rate vs Price Value Rating (line), Top 1000 Revenue Listing (review-focused table), Reviews Frequency by Borough (table).
Filters / Slicers	Same global slicer panel as Overview.
Questions Answered	How satisfied are guests across categories? How does review activity/recency vary by borough? Is there a relationship between price and rating?
Business Value	Supports quality monitoring and highlights boroughs or segments where review engagement is weakening.

9. Detailed Dashboard Analysis

9.1 Overview Page

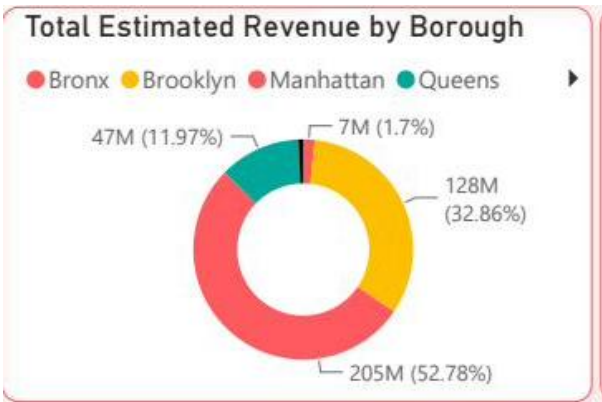


Overview page — full dashboard screenshot.

Purpose: Establish the top-line size and revenue distribution of the marketplace and show how pricing varies by borough, room type, and bathroom privacy.

KPI Analysis: Total Estimated Revenue (**389M**) and Total Listing (**30K**) imply roughly **\$12,800** revenue per listing, while the explicit Revenue per Available KPI (**18.11K**) is higher — consistent with revenue concentrating among the **22K** Active Listings rather than spreading across all **30K**.

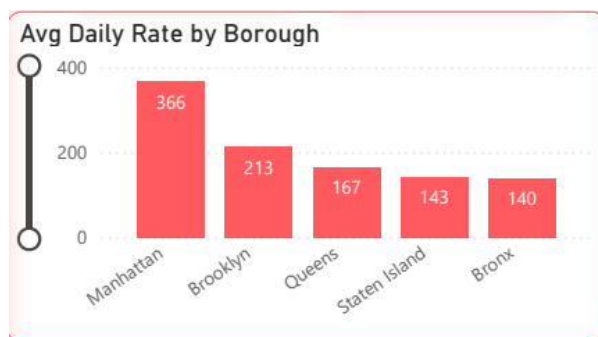
Visual: **Total Estimated Revenue by Borough**



Observation: Manhattan **205M (52.78%)**, Brooklyn **128M (32.86%)**, Queens **47M (11.97%)**, Bronx **7M (1.7%)**.

Insight: Over **85%** of revenue comes from just two boroughs — Manhattan and Brooklyn are the commercially dominant boroughs; Queens, Bronx, and Staten Island are a small share of current revenue.

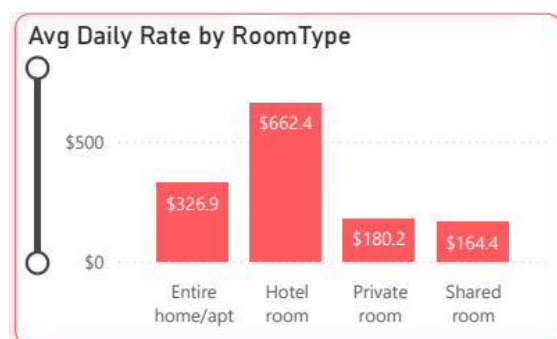
Visual: **Avg Daily Rate by Borough**



Observation: Manhattan leads at **\$366**, over **70%** above Brooklyn (**\$213**); Queens (**\$167**), Staten Island (**\$143**), Bronx (**\$140**) cluster lower.

Insight: Manhattan's revenue dominance is driven by a sustained price premium, not just listing volume.

Visual: Avg Daily Rate by RoomType



Observation: Hotel room is highest at **\$662.4**, then Entire home/apt (**\$326.9**), Private room (**\$180.2**), Shared room (**\$164.4**).

Insight: Hotel-style listings, while a small share of supply, command a large per-unit price premium.

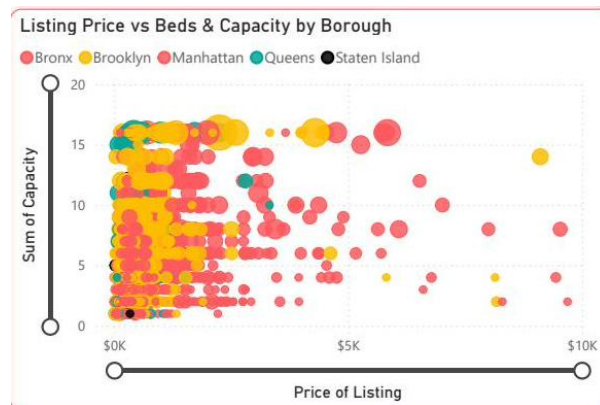
Visual: Avg Daily Rate by Bathroom Privacy



Observation: “Unknown” privacy listings price highest (**\$332.4**), ahead of Private (**\$268.9**) and Shared (**\$116.4**).

Insight: Counter-intuitive: this likely reflects a data-completeness gap among higher-end listings rather than a genuine pricing driver — worth investigating before using bathroom privacy in pricing decisions.

Visual: Listing Price vs Beds & Capacity by Borough



Observation: Most listings cluster under **\$2K** price and **0–15** capacity; a small set of outliers extend toward **\$10K**, mostly Brooklyn/Manhattan.

Insight: The distribution is heavily right-skewed — a handful of premium, high-capacity listings should be handled separately in any pricing model.

Visual: Top 1000 Revenue Listing (table)

ListingKey	Total Est Revenue	Price	Est Beds	Capacity	Overall [®]
1	\$6,838.0	\$113.97	2	2	
2	\$21,109.0	\$117.27	2	3	
3	\$4,804.0	\$80.06	1	1	
4	\$9,260.0	\$77.17	1	1	
Total	\$38,92,98,622.0	\$269.49	1.6914833188169804	86625	

Observation: Total row reconciles to marketplace totals: **\$389,298,622** revenue, **\$269.49** avg price, **1.69** avg est. beds, **86,625** total capacity.

Insight: This table's Total row is an audit trail linking listing-level detail back to the top-line KPIs.

9.2 Host Page



Host page — full dashboard screenshot.

Purpose: Examine how listing supply is distributed across the host base, and how tenure and portfolio size relate to revenue and room-type mix.

KPI Analysis: With **16,442** hosts averaging **4.90** listings each, but only **16.0%** Host Active Listings and a **19.8%** Superhost rate, the host base is large and only partially active or elite.

Visual: Top 10 Host Performance by Total Listing



Observation: A named host (“David”) shows a bar far above the next-highest names, declining steadily thereafter.

Insight: This ranking is by host first name, likely aggregating several distinct hosts who share that name — read as name-level concentration, not necessarily one dominant host, pending HostID-level confirmation.

Visual: **Host Portfolio**



Observation: **71.04%** of hosts (**11.68K**) are Single-Listing, **23.64%** (**3.89K**) Small Portfolio, the rest Medium/Large.

Insight: Roughly seven in ten hosts are casual, single-property operators, not professional multi-listing businesses — relevant to how host support should be designed.

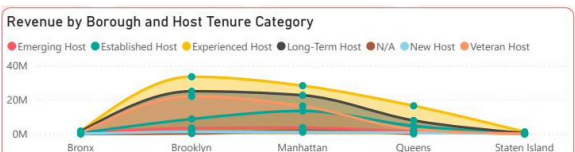
Visual: **Top 10 Host by Listing (by borough)**

Top 10 Host by Listing		
Borough	Sum of TotalListings	Hosts
Bronx	1262	
Brooklyn	13760	
Manhattan	55192	
Queens	9905	
Arverne	118	
Total	80552	

Observation: Manhattan **55,192**, Brooklyn **13,760**, Queens **9,905**, Bronx **1,262**; total **80,552** — above the **30,260** listing count.

Insight: The model diagram (Section 4) confirms dim_host is listing-grain and 1:1 with fact_listing, so a plain Sum of TotalListings by Borough counts each multi-listing host once per NYC listing they own, inflating the total above both the true host count (**16,442**) and listing count. Read **80,552** as a supply-weighted concentration figure, not a literal count.

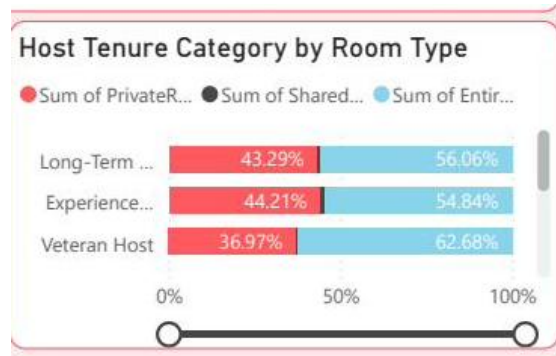
Visual: **Revenue by Borough and Host Tenure Category**



Observation: Revenue peaks for Established/Experienced hosts in Brooklyn and Manhattan (**20–40M** range), tapering sharply toward Queens and Staten Island.

Insight: Host-tenure incentive programs would have the largest revenue impact if focused on mid-tenure hosts in the two dominant boroughs.

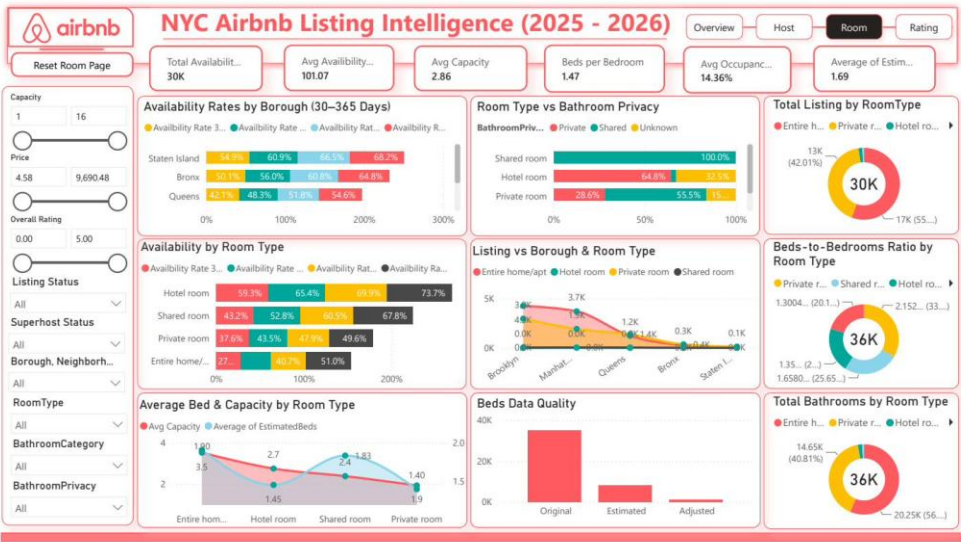
Visual: Host Tenure Category by Room Type



Observation: Veteran Hosts skew **62.68%** Entire home/apt vs **36.97%** Private room; Long-Term Hosts are more balanced (**56.06%** / **43.29%**).

Insight: Longer-tenured hosts increasingly favor whole-unit rentals — possibly reflecting professionalisation as hosts gain experience.

9.3 Room Page

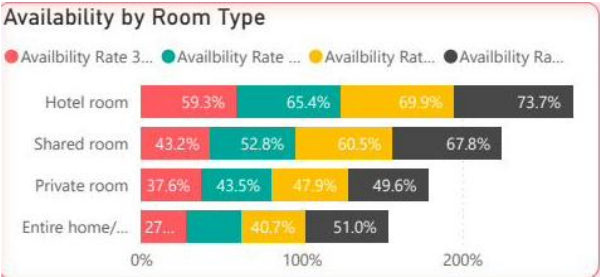


Room page — full dashboard screenshot.

Purpose: Analyze how room type, bathroom configuration, and bed/capacity data relate to availability.

KPI Analysis: Avg Capacity of **2.86**, Beds per Bedroom of **1.47**, and Average Estimated Beds of **1.69** point to a marketplace of predominantly small-to-mid-sized units.

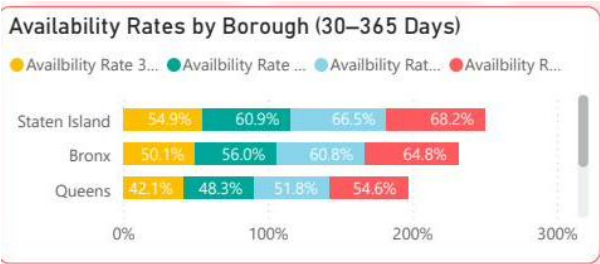
Visual: **Availability by Room Type**



Observation: Entire home/apt has the lowest availability across windows (**27–51%**); Hotel room the highest (**59.3–73.7%**).

Insight: Entire-unit listings are booked/blocked more often, consistent with their higher listing share and pricing — likely reflecting stronger demand for private, whole-unit stays.

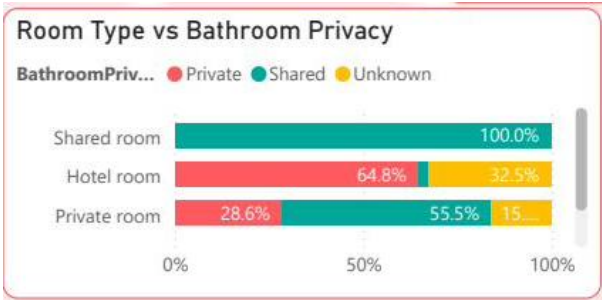
Visual: **Availability Rates by Borough (30–365 Days)**



Observation: Of the boroughs visible, Staten Island is highest (**54.9–68.2%**), then Bronx (**50.1–64.8%**), then Queens (**42.1–54.6%**).

Insight: Lower-revenue boroughs (Section 9.1) show comparatively higher, more open availability — consistent with weaker demand pressure rather than confirmed oversupply.

Visual: Room Type vs Bathroom Privacy



Observation: Private room listings are **55.5%** Shared bathroom and only **28.6%** Private; Hotel room is **64.8%** Private.

Insight: A majority of “Private room” listings do not actually have a private bathroom — a naming/expectation gap worth disclosing clearly to guests.

Visual: Total Listing by RoomType



Observation: Entire home/apt is **55% (17K)** of listings, Private room **42.01% (13K)**, Hotel room a small remainder.

Insight: Supply is concentrated in two room types — Hotel room's high price (Section 9.1) applies to a small, premium niche.

Visual: Beds Data Quality



Observation: "Original" bed counts far outnumber "Estimated", with "Adjusted" smallest.

Insight: Bed-count imputation (Section 5) was applied only where necessary, preserving original values as the primary source.

Visual: Total Bathrooms by Room Type



Observation: Entire home/apt is 56% (20.25K) and Private room 40.81% (14.65K) of a 36K total — above the 30K listing count.

Insight: A meaningful share of Entire home/apt units have more than one bathroom, likely a distinct, higher-capacity segment.

9.4 Rating Page



Rating page — full dashboard screenshot.

Purpose: Assess guest satisfaction across rating sub-categories and review-engagement patterns by borough.

KPI Analysis: All rating sub-categories sit in a tight, high band (4.64–4.84 of 5) — Check-In Rating highest (4.84), Value Rating lowest (4.64).

Visual: Reviews Frequency by Borough

Reviews Frequency by Borough				
Borough	Total Review	L30Days	L12Months	Reviews/Month %
Bronx	30,673	333	3,470	79.5%
Brooklyn	4,08,415	3,740	46,424	80.4%
Manhattan	3,59,922	4,507	59,448	69.3%
Queens	1,68,042	1,905	23,214	95.1%
Staten Island	13,500	87	1,159	86.6%
Total	9,80,552	10,572	1,33,715	78.2%

Observation: Queens highest Reviews/Month at 95.1%, then Staten Island (86.6%), Brooklyn (80.4%), Bronx (79.5%), Manhattan lowest (69.3%); total 78.2%.

Insight: Manhattan's high price and revenue do not translate into the highest review-engagement rate — worth investigating guest mix or stay length.

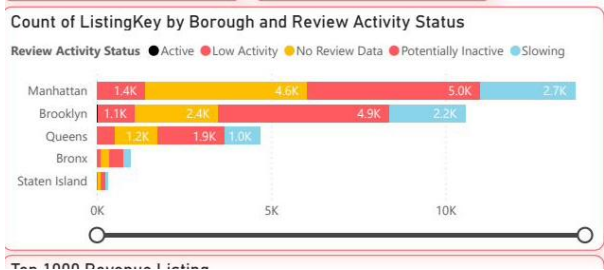
Visual: Top 1000 Revenue Listing / Review Table (by Borough)

Top 1000 Revenue Listing				
Borough	TotalReviews	Days Since Last Review	Review Lifespan Year	Review A
Staten Island	13,500	178594	837	Low Activ
Bronx	30,673	562704	2057	Low Activ
Queens	1,68,042	2794914	9443	Active
Manhattan	3,59,922	9304325	25243	Active
Brooklyn	4,08,415	9021438	26044	Active
Total	9,80,552	21861975	63624	Active

Observation: Brooklyn and Manhattan show the largest cumulative review totals and are both classified Active; Staten Island and Bronx are Low Activity.

Insight: The two top-revenue boroughs are also the most reviewed and active, reinforcing them as the marketplace's most mature segments.

Visual: Count of ListingKey by Borough and Review Activity Status



Observation: Manhattan and Brooklyn show the largest “No Review Data” segments (roughly 5.0K and 4.9K) alongside their Active segments.

Insight: Even in the most mature boroughs, a substantial share of listings carry no review data — important context since rating KPIs only reflect reviewed listings.

Visual: Daily Rate vs Price Value Rating



Observation: Manhattan reaches the highest prices (up to ~\$350–380) with the most volatility across the 1–5 rating axis; other boroughs cluster lower.

Insight: Price does not rise smoothly with value rating in any borough — rating alone is not a reliable predictor of price here.

10. Exploratory Data Analysis

This section condenses cross-cutting patterns from Section 9 into five themes (full visual evidence is in Section 9).

Market & Host Base: 30K total listings, 22K (~73%) active, 389M revenue, 14.4% avg occupancy, 4.77 avg rating. 16,442 hosts average 4.90 listings each, but 71.04% hold only one listing — a casual, long-tail host base with near-universal verification (98.9%) but minority Superhost status (19.8%).

Geographic: Manhattan leads on rate (\$366) and revenue share (52.78%), Brooklyn second (\$213, 32.86%); Queens/Staten Island/Bronx are a lower-price, lower-revenue tier (together under 15% of revenue). Availability runs inversely — Staten Island and Bronx show the highest availability among boroughs measured.

Room Type & Pricing: Entire home/apt is the largest segment (55%) and second-highest priced (\$326.9); Hotel room is a small niche but highest priced (\$662.4, roughly 4x Shared room). Average price overall is \$269.49. The Bathroom Privacy “Unknown” category pricing highest is a likely data-quality artifact (Section 9.1), not a confirmed driver.

Reviews: 980,552 total reviews, dominated by Brooklyn (408,415) and Manhattan (359,922). Reviews/Month is highest in Queens (95.1%) and lowest in Manhattan (69.3%) — volume and engagement consistency do not move together.

Availability: Avg availability 101.07, avg occupancy 14.36% (Room page). Highest for Hotel room and in Staten Island/Bronx; lowest for Entire home/apt — consistent across Overview, Host, and Room pages.

11. Key Insights & Findings

Insight 1: Manhattan's revenue dominance is price-driven, not volume-driven

Finding: Manhattan generates 52.78% of revenue (205M) with an average daily rate of \$366 — over 70% above the next borough.

Evidence: Avg Daily Rate by Borough and Total Estimated Revenue by Borough, Overview page.

Interpretation & Implication: Manhattan's revenue share is explained by a sustained price premium, not a proportionally larger listing share. Pricing/premium positioning matters more than raw supply growth for revenue in Manhattan; other boroughs shouldn't assume similar per-unit revenue without addressing the price gap.

Insight 2: The host base is dominated by casual, single-listing hosts

Finding: 71.04% of hosts (11.68K) hold only one listing; average listings-per-host is 4.90, pulled up by a smaller multi-listing group.

Evidence: Host Portfolio donut, Host page.

Interpretation & Implication: Most hosts are casual, small-scale participants rather than professional operators. Host-support programs should target this large, low-portfolio segment primarily, with a separate high-touch track for the smaller professional cohort.

Insight 3: “Private room” listings frequently do not have a private bathroom

Finding: 55.5% of Private room listings have a Shared bathroom, versus only 28.6% Private.

Evidence: Room Type vs Bathroom Privacy, Room page.

Interpretation & Implication: Room-type label alone does not reliably communicate bathroom privacy to guests. This is a guest-experience risk — mismatched expectations could lower Value/Cleanliness ratings if not clearly disclosed at booking.

Insight 4: Availability and revenue move in opposite directions across boroughs

Finding: Manhattan and Brooklyn (highest revenue) are also the most review-Active boroughs, while Staten Island and Bronx (highest availability) are Low Activity.

Evidence: Availability Rates by Borough cross-referenced with revenue and review-activity tables.

Interpretation & Implication: Higher availability in lower-revenue boroughs suggests weaker demand pressure rather than confirmed oversupply. Demand-side interventions in Staten Island/Bronx may have more room to improve occupancy than new host acquisition.

Insight 5: Superhost status and active engagement are minority conditions

Finding: Only **19.8%** of hosts are Superhosts and **16.0%** of listings are active, despite **98.9%** identity verification.

Evidence: Super Host Rate, Host Active Listings, Identity Verification KPIs, Host page.

Interpretation & Implication: Trust/safety compliance is high, but ongoing engagement and top-tier status are comparatively rare. There may be room to grow Superhost and active-listing rates among an already-verified host base.

12. Cross-Visual Insights

Looking across the four dashboard pages together surfaces relationships not visible from any single page:

- High price + high revenue + lower availability: Manhattan combines the highest average daily rate, the highest revenue share, and — among the boroughs shown on the Room page — comparatively tighter availability, suggesting price and demand pressure are associated with each other in this market, though the underlying booking/occupancy data needed to confirm true demand-driven scarcity was not provided.
- High listing volume + lower relative price per unit: Entire home/apt is both the largest room-type segment and mid-tier priced (behind Hotel room), while commanding the lowest availability of any room type — consistent with high demand for whole-unit privacy outweighing its comparatively larger supply base.
- Host verification + Superhost status appear largely decoupled: 98.9% of hosts are identity-verified, but only 19.8% are Superhosts, indicating verification (a baseline trust requirement) and Superhost status (a performance-based designation) are not closely linked in this dataset, though a direct correlation statistic was not available.
- Review volume and review-rate consistency diverge by borough: Manhattan and Brooklyn have the highest raw review counts but Manhattan has the lowest Reviews/Month percentage, suggesting raw scale and engagement consistency are separate signals that should be tracked independently rather than assumed to move together.
- Bathroom-privacy pricing anomaly may be a data-quality signal rather than a market pattern: the “Unknown” bathroom-privacy category showing the highest average price is unusual and is more likely to reflect a gap in how certain (possibly higher-end) listings report bathroom details than a genuine guest willingness to pay more for undisclosed privacy status.

These are described as associations observed alongside one another in the dashboard, not as confirmed causal relationships — the data available does not support causal claims (e.g., that lower availability causes higher price, or that price causes lower review frequency).

13. Business Recommendations

Airbnb Marketplace Team

Finding	Business Interpretation	Recommendation	Expected Benefit
Manhattan and Brooklyn generate 85.6% of revenue	Marketplace commercial performance is concentrated in two boroughs	Prioritize retention and quality initiatives in Manhattan/Brooklyn while running targeted host-acquisition or demand-generation pilots in Queens, Staten Island, and Bronx	Protects the revenue core while testing lower-cost paths to diversify revenue geography
Staten Island and Bronx show higher availability but lower review activity	These boroughs may be under-demanded rather than under-supplied	Focus marketing/demand-side spend (rather than new host recruitment) on these boroughs	More efficient use of growth budget by matching the intervention to the actual gap (demand, not supply)
“Private room” listings often have shared bathrooms	Listing taxonomy may create guest expectation mismatches	Introduce or reinforce mandatory bathroom-privacy disclosure at the listing level, independent of room-type label	Reduced guest dissatisfaction and potential uplift in Value/Cleanliness-related ratings

Hosts

Finding	Business Interpretation	Recommendation	Expected Benefit
Hotel room listings command the highest average daily rate (\$662.4) on a small supply base	A premium, under-supplied niche exists within the marketplace	Hosts with suitable properties could consider positioning listings as hotel-room-style offerings where appropriate	Potential access to a higher price tier without necessarily expanding total unit count
Entire home/apt has the lowest availability across room types	Whole-unit listings are in comparatively higher demand	Hosts operating Private/Shared rooms could evaluate whether converting to whole-unit rental (where feasible) improves utilization	Improved occupancy for hosts able to make this transition
Value Rating (4.64) is the lowest-rated sub-category despite high overall satisfaction	Guests are marginally less satisfied with price-to-value perception than with other service dimensions	Hosts, especially in higher-priced boroughs, should review pricing relative to amenities/condition	Incremental improvement in guest satisfaction and repeat-booking likelihood

Business / Operations Team

Finding	Business Interpretation	Recommendation	Expected Benefit
71% of hosts are single-listing operators	The host base is predominantly casual rather than professional	Track host segment (single vs. multi-listing) as a standing operational KPI to tailor support resources	More efficient allocation of host-success resources across a large, heterogeneous host base
Superhost rate (19.8%) is well below identity-verification rate (98.9%)	There may be headroom to convert verified hosts into higher-performing, Superhost-status hosts	Monitor and report on the verified-to-Superhost conversion funnel over time	A clearer view of where host-quality investment would have the most leverage
Three related rating figures (4.77 / 4.75 / 4.73) use three distinct, individually valid DAX scopes (Section 7)	Each figure answers a slightly different question (reviewed listings only vs. all listings, direct rating vs. composite of six sub-ratings), which is not obvious to a dashboard viewer	Add a short tooltip or subtitle to each rating card/gauge stating its exact scope (e.g., "Listings with 5+ reviews")	Prevents viewers from mistakenly treating the three figures as a data error or picking the wrong one for a decision

14. Dashboard UX/UI Evaluation

Visual Hierarchy

Each page places its headline KPI cards in a consistent horizontal strip at the top, immediately below the page-navigation tabs, which gives users a stable, predictable orientation point before drilling into supporting visuals below.

Interactivity

A single, consistent slicer panel (Capacity, Price, Overall Rating, Listing Status, Superhost Status, Borough/Neighborhood, RoomType, BathroomCategory, BathroomPrivacy) is present on all four pages, enabling cross-page filtering without needing to relearn controls. Each page also includes a dedicated "Reset [Page] Page" button, a good practice for exploratory dashboards. No drill-through pages, tooltips, or bookmarks were visible in the provided screenshots, so their presence/absence cannot be further evaluated beyond what is shown.

Visualization Selection

Most chart types are well matched to their questions: donuts for share-of-total (revenue, room type, host portfolio), bar charts for categorical comparison (average daily rate by borough/room type), and a scatter plot for the two-variable price-vs-capacity relationship. Some visuals — such as the Top 10 Host Performance by Total Listing combination chart and the Availability by Room Type stacked-percentage bars — are visually dense and would benefit from simplification (see Section 15).

Usability

Labeling is generally clear, though several KPI card titles are truncated in the screenshots (e.g., "Revenue per Av...", "Avg Occupanc...", "Host Active Lis..."), which would affect readability at the dashboard's native resolution unless

card widths are adjusted. Consistent axis formatting (percentages, currency, K/M suffixes) is used throughout, aiding cross-visual comparison.

Professionalism

The dashboard maintains a consistent Airbnb-inspired color palette and layout structure across all four pages, with aligned KPI card rows and a repeated navigation bar — this consistency supports a professional, cohesive storytelling experience appropriate for a portfolio piece.

15. Data-Driven Recommendations for Dashboard Improvement

- Fix truncated KPI card titles (e.g., “Revenue per Av...”, “Avg Occupanc...”) by widening cards or shortening label text so every KPI is fully readable without hovering.
- Reconcile the Overall Rating discrepancy (4.77 KPI card vs. 4.73/4.75 gauges) by standardizing the underlying DAX aggregation logic across all rating measures.
- Add a drill-through page from the “Top 10 Host Performance by Total Listing” visual to a single-host detail view, to clarify whether name-level concentration (e.g., “David”) reflects one dominant host or several distinct hosts sharing a first name.
- Add tooltip pages on borough-level visuals (e.g., Total Estimated Revenue by Borough) showing supporting metrics like listing count and average rating on hover, reducing the need to flip between pages to cross-reference.
- Introduce a Key Influencers visual on the Rating page to identify which factors (borough, room type, price, host tenure) most strongly associate with lower Value or Overall ratings.
- Add a Decomposition Tree on the Overview page to let users break down Total Estimated Revenue interactively by Borough → Room Type → Bathroom Privacy without needing multiple separate charts.
- Standardize borough lists in scrollable visuals (e.g., Availability Rates by Borough only shows 3 of 5 boroughs before scrolling) so all boroughs are visible without interaction, or add a clear “scroll for more” visual cue.
- Apply conditional formatting to the Reviews Frequency by Borough and Top 1000 Revenue Listing tables to visually flag the strongest/weakest performing rows at a glance.

16. Limitations & Assumptions

Data Limitations

- The dataset appears to represent a single time snapshot (2025–2026) rather than a historical time series, limiting trend analysis over multiple years.
- `gold_dim_host` is confirmed (via the Power BI model diagram, Section 4) to be listing-grain and related 1:1 with `fact_listing` — a host with multiple listings appears once per listing, each time repeating that host's portfolio attributes. Any Sum-based aggregation of host-level fields (e.g., `TotalListings`) by another dimension will overcount multi-listing hosts; `DISTINCTCOUNT`-based measures are unaffected.
- Host-level `TotalListings` figures are self-reported platform-wide counts (from the source listings data) and may include listings outside NYC or outside the dataset's collection window, so they should not be equated with the number of NYC listings this project has data for.

- The Power Query M script includes Table.Distinct de-duplication logic on gold.dim_host by HostID, but this is not reflected in the 1:1, listing-grain relationship shown in the actual Power BI model diagram — an unresolved discrepancy between the provided M code and the built model that the project owner should reconcile (see Information Required).
- dim_date reflects a single price-quote check-in/check-out window per listing (from price_quote_checkin_date / price_quote_checkout_date), not a history of actual guest bookings — it cannot be used to analyze booking patterns over time.
- The exact upstream raw data source (e.g., which public NYC Airbnb export and vintage) was not specified in the provided materials; it is only known that raw CSV files were loaded into a SQL Server dbo staging schema before entering the bronze layer.

Analytical Limitations

- This is a descriptive analysis; no predictive or statistical modeling (e.g., regression, forecasting) is present in the reviewed materials.
- Correlation shown across visuals (e.g., price vs. availability, price vs. review activity) does not establish causation.
- Review count and review frequency are proxies for guest engagement, not direct measures of actual bookings.
- Availability (30/60/90/365-day windows) reflects open calendar days, not confirmed occupancy; the Avg Occupancy KPI is a separate estimated measure and the two should not be conflated.
- Estimated Revenue is a modeled/derived figure (as its name indicates) and not necessarily equal to realized, collected revenue.

Dashboard Limitations

- The dashboard reflects a static, non-real-time dataset extract; there is no evidence of a live refresh connection in the provided materials.
- No external benchmarking data (e.g., hotel pricing, tourism volume) is incorporated, limiting context for whether observed patterns are Airbnb-specific or reflect broader NYC travel demand.

17. Conclusion

This project analyzed a 30,260-listing, eight-table NYC Airbnb gold-layer dataset through a four-page Power BI dashboard covering marketplace overview, host structure, room/availability configuration, and guest rating performance. The dashboard reveals a marketplace whose revenue is heavily concentrated in Manhattan and Brooklyn, driven primarily by price rather than listing volume; a host base dominated by casual, single-listing participants alongside a smaller multi-listing cohort; a room-type mix where whole-unit listings dominate supply but show the tightest availability; and a consistently high, though not perfectly uniform, guest satisfaction profile across rating categories and boroughs.

The dashboard provides tangible business value: it allows stakeholders to see, in a few clicks, where revenue is concentrated, which host segments matter most, where availability and demand appear mismatched, and where guest experience risks (such as the room-type/bathroom-privacy naming gap) may exist. Marketplace, host-success, and operations teams could each use this tool as a starting point for the specific recommendations

outlined in Section 13, while the flagged data-quality and modeling nuances (Sections 9.2, 15, 16) provide a clear, prioritized punch list for strengthening the underlying model before extending it further.

18. Future Scope

The following are realistic extensions to this project and are explicitly future possibilities, not existing capabilities of the current dashboard:

Advanced Power BI

- Forecasting visuals to project revenue or occupancy trends if multi-period historical data becomes available.
- What-if parameters to let users simulate the revenue impact of price or occupancy changes by borough.
- Key Influencers and Decomposition Tree visuals for deeper diagnostic analysis of rating and revenue drivers.
- Drill-through pages from borough/host summary visuals to individual listing or host detail views.

Advanced Analytics

- Price-prediction modeling based on room type, location, capacity, and bathroom configuration.
- Booking-demand prediction using historical occupancy patterns, if booking-level transaction data is obtained.
- Host segmentation using clustering techniques beyond the current tenure/portfolio-size categories.
- Anomaly detection to systematically flag data-quality issues such as the bathroom-privacy pricing anomaly identified in this report.

External Data Integration

- NYC tourism volume and seasonality data to contextualize occupancy and availability patterns.
- Local events calendars to explain short-term demand spikes.
- Comparable hotel pricing data to benchmark Airbnb pricing against the broader accommodation market.
- Public transportation/accessibility data to explore whether proximity to transit correlates with pricing or occupancy.

19. Technical Appendix

19.1 Data Dictionary

Field	Table	Description	Data Type
ListingKey	All tables	Surrogate key linking all fact/dimension tables	Integer
ListingID	fact_listing	Native platform listing identifier	Integer
HostID	fact_listing dim_host	/ Host identifier	Integer

Field	Table	Description	Data Type
Price	fact_listing	Nightly listing price	Currency
EstRevenue	fact_listing	Estimated revenue for the listing	Currency
EstOccupancy	fact_listing	Estimated occupancy measure	Numeric
Avail30/60/90/365Days	fact_listing	Days available within each rolling window	Integer
TotalReviews / ReviewsPerMonth	fact_listing	Review volume and frequency measures	Integer / Decimal
Rating / AccuracyRating / CleanlinessRating / CheckInRating / CommunicationRating / LocationRating / ValueRating	fact_listing	Overall and sub-category guest ratings (0–5)	Decimal
HostName / IsSuperhost / IsIdentityVerified	dim_host	Host identity and status flags	Text / Boolean
TotalListings / ActiveListings / EntireHomeListings / PrivateRoomListings / SharedRoomListings	dim_host	Host portfolio composition (repeated per listing row)	Integer
UserYears/Months / HostYears/Months / TotalUserMonths / TotalHostMonths	dim_host	Host tenure measures	Integer
Borough / Neighborhood / Latitude / Longitude	dim_location	Geographic classification and coordinates	Text / Decimal
RoomType / Capacity / Bedrooms / Beds / EstimatedBeds / BedDataQuality	dim_room	Room configuration and bed-count quality flag	Text / Integer
BathroomType / BathroomCount / BathroomPrivacy / BathroomCategory	dim_bathroom	Bathroom configuration	Text / Integer
HasAvailability / AvailYearEnd	dim_avability	Availability status flags	Boolean / Integer
FirstReview / LastReview	dim_review	Review date range for the listing	Date
CheckInDate / CheckOutDate	dim_date	Price-quote check-in/check-out window (not booking history — see Section 3.3)	Date

19.2 DAX Measures

The following measures and calculated columns are defined in dax_measures_columns_script.txt and power the dashboard's KPI cards and derived categories.

Calculated Columns

Name	Formula (summarized)	Used For
Listing Status	IF(ISBLANK([Price]), "Inactive", "Active")	Active/Inactive Listing counts
Host Tenure Category	SWITCH on TotalHostMonths ranges: 0–6 New, 7–12 Emerging, 13–24 Established, 25–60 Experienced, 61–120 Long-Term, 120+ Veteran, blank N/A	Host Tenure Category visuals (Section 9.2)

Key Measures (grouped by theme)

Theme	Measure	Formula (summarized)
Availability	Total Listing Availability	DISTINCTCOUNT(ListingKey) where HasAvailability = 1
Availability	Avg 30/60/90/365 Days Availability	AVERAGE(Avail30/60/90/365Days)
Availability	Availability Rate 30/60/90/365 Days	DIVIDE(Avg Availability X Days, X, 0)
Availability	Availability Coverage %	DIVIDE(Total Availability Listing, Total Listing, 0)
Availability	Average Availability by Horizon	SWITCH(SELECTEDVALUE(Horizon), maps to the matching Avg Availability X Days measure)
Rating	Average Rating Across Categories	AVERAGEX over all listings of (Value+Accuracy+CheckIn+Cleanliness+Communication+Location)/6
Rating	Avg Overall Rating	CALCULATE(AVERAGE(Rating), TotalReviews >= 5)
Rating	Average Review Lifespan Years	AVERAGE('gold dim_review'[Review Lifespan Year])
Rating	Total Reviews / Listing Review Frequency	SUM(TotalReviews); DIVIDE(Total Reviews, Total Listing, 0)
Revenue	Total Estimated Revenue	SUM(EstRevenue)
Revenue	Revenue per Available Listing	DIVIDE(Total Estimated Revenue, Active Listing, 0)
Revenue	Neighborhood Revenue Rank	RANKX of Total Estimated Revenue within each Borough, using REMOVEFILTERS + CALCULATETABLE
Listing Capacity /	Total Listing / Listing Count	COUNT(ListingID); DISTINCTCOUNT(ListingKey)
Listing Capacity /	Avg Capacity / Avg Daily Rate / Beds per Bedroom	AVERAGE(Capacity); AVERAGE(Price); DIVIDE(SUM(EstimatedBeds), SUM(Bedrooms))

Theme	Measure	Formula (summarized)
Listing Capacity /	Avg Occupancy Rating	DIVIDE(AVERAGE(EstOccupancy), 365, 0)
Host	Average Listings per Host / Avg host listing	DIVIDE(SUM(TotalListings), COUNT(HostID), 0); AVERAGE(TotalListings)
Host	Host Active Listing Rate	DIVIDE(MAX(ActiveListings), MAX(TotalListings), 0)
Host	Host Listing Rank	RANKX(ALL(HostID), Total Listing, DESC, Dense)
Host	Total Host count / Super Host Rate / Total SuperHost	DISTINCTCOUNT(HostID); DIVIDE(Total SuperHost, Total Host count, 0); COUNT where IsSuperhost = 1
Host	Identity Verification Rate / Total Identity Verified	DIVIDE(Total Identity Verified, Total Host count, 0); COUNT where IsIdentityVerified = 1

Not all dashboard elements have a corresponding measure in this script — notably, “Review Activity Status” (Active/Slowing/Low Activity/Potentially Inactive/No Review Data, used on the Rating page) does not appear in the provided DAX file, so its exact logic remains unconfirmed. Bathroom Category/Privacy and Beds Data Quality are confirmed to be computed in SQL (Section 5), not DAX.

19.3 SQL Queries

Full scripts were provided for all three warehouse layers. Representative highlights are summarized below; see Sections 4–5 for how each is used analytically.

Script	Layer	Key Logic
main_ddl.sql	Setup	Drops/recreates the NYC_Airbnb_DW database and creates the bronze, silver, and gold schemas.
bronze_listing_ddl.sql / bronze_..._stored_procedure.sql	Bronze	Creates bronze.nyc_airbnb_listing with every column as NVARCHAR(100); bronze.load_bronze loads it from dbo.nyc_airbnb_listing via TRY_CAST.
silver_listing_ddl.sql / silver_..._stored_procedure.sql	Silver	Creates the typed silver.NYC_airbnb_listing table; silver.load_silver truncates and reloads it from Bronze with NULLIF/TRY_CAST/TRY_CONVERT null handling, a ROW_NUMBER()-based ListingKey, host-tenure month derivation, bathroom text parsing, and price cleaning (see Section 5 for exact logic).
gold_stored_procedure.sql	Gold	Defines gold.dim_host, gold.dim_location, gold.dim_room, gold.dim_bathroom, gold.dim_availability, gold.dim_review, gold.dim_date, and gold.fact_listing as views over silver.NYC_airbnb_listing, applying boolean-flag standardization, beds imputation, and bathroom categorization.

19.4 Data Model

Confirmed directly from the Power BI model-view screenshot (Figure 1, Section 4): gold.fact_listing is related 1:1 on ListingKey to all seven gold dimension views (dim_host, dim_review, dim_bathroom, dim_room, dim_location,

dim_availability, dim_date), each holding the same 30,260-row listing grain as the fact table — including dim_host, which is not de-duplicated in the built model despite the de-duplication logic present in the Power Query M script. The dim_room relationship is configured for bidirectional cross-filtering; the rest use the default single direction. Two disconnected calculated tables (Key Measures, Availability Horizon) and one hidden measure-organization table (Measure_Table) complete the model, matching their definitions in dax_measures_columns_script.txt exactly. All primary/foreign keys are hidden from the report view.

19.5 Dashboard Screenshots Reference

Page #	Page Name	Primary Focus
1	Overview	Revenue, pricing, and marketplace-level KPIs
2	Host	Host concentration, tenure, and portfolio structure
3	Room	Room type, bathroom configuration, and availability
4	Rating	Guest satisfaction and review engagement